

MODULE 01 / 40

Northern Cardinal

*Cardinalis cardinalis***QUICK STATS**

Length	8.3 – 9.3 in (21 – 23.5 cm)
Wingspan	9.8 – 12.2 in (25 – 31 cm)
Weight	1.5 – 1.7 oz (42 – 48 g)
Lifespan	Up to 15 years (avg. 3 in wild)

PRESENTATION

The Northern Cardinal is one of the most iconic and instantly recognizable birds visiting North American gardens year-round. The brilliant scarlet male, with his dramatic crest and orange-red bill, is among the most photographed birds on the continent. The female, dressed in warm buff-brown with reddish tinges on her crest, wings and tail, is no less elegant. Both sexes remain on their territory throughout winter, bringing color and song to snow-covered gardens when most other species have migrated south.

HABITAT & RANGE

The Northern Cardinal thrives across the eastern half of North America, from southern Maine and the Great Lakes south through Florida, and westward to the Great Plains of Texas and Oklahoma. Populations are also well-established in Arizona, New Mexico, and parts of California. Southern Ontario and Quebec host significant breeding populations. The species has expanded its range northward over the past century, likely aided by the proliferation of bird feeders and milder winters. It favors woodland edges, dense shrubby thickets, hedgerows, suburban gardens, and parks with mature trees and dense understory cover.

DIET & FEEDING

- Black-oil sunflower seeds — the undisputed favorite
- Safflower seeds — especially attractive to cardinals
- White millet and cracked corn
- Wild berries: dogwood, holly, elderberry, sumac
- Insects and larvae (essential protein during breeding season)
- Fruit: grapes, mulberries, raspberries

BEHAVIOR & SOCIAL LIFE

Cardinals are territorial songbirds that maintain year-round home ranges of roughly 2–4 acres. Males fiercely defend their territory through song and, famously, by attacking their own reflection in windows — mistaking it for a rival. They are monogamous and pairs often remain together through multiple breeding seasons. Outside the breeding season, cardinals may gather in loose flocks of up to 70 individuals at abundant food sources. Males frequently feed females during courtship — a behavior called mate-feeding — passing seeds beak-to-beak.

NESTING & BREEDING

Breeding season runs from March through September, with females typically raising 2 to 4 broods per year — one of the highest rates among North American songbirds. The female constructs a cup-shaped nest over 3–9 days using twigs, bark strips, grasses and leaves, lined with fine grasses and hair. Nests are built 1–15 feet above ground in dense shrubs, vines or low trees. Each clutch contains 2–5 eggs (usually 3), pale whitish or greenish with brown speckles. Incubation lasts 11–13 days, performed almost exclusively by the female. Nestlings fledge at 9–11 days and the male takes over their care while the female begins a new nest.

SONG & COMMUNICATION

The Northern Cardinal is renowned for its rich, loud whistled song — a series of clear slurred notes often described as "cheer-cheer-cheer" or "birdy-birdy-birdy." Both males and females sing, which is unusual among North American songbirds. The female often sings from the nest, possibly communicating food needs to her mate. Cardinals have a repertoire of 16 or more distinct song patterns. Their sharp metallic "chip" call is an alarm note used to warn of predators. Song peaks in late winter and early spring as males establish territories.

CONSERVATION STATUS

IUCN Red List:	Least Concern	Population trend:	Stable / Increasing
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With an estimated population of over 100 million individuals, the Northern Cardinal is one of the most abundant songbirds in North America. Its range has expanded northward significantly since the 1900s. It is the official state bird of 7 US states — more than any other species.

FEEDER TIPS

- Best feeder type: hopper feeder or large platform feeder
- Preferred seeds: safflower (squirrels tend to avoid it) and black-oil sunflower
- Cardinals feed most actively at dawn and dusk
- Place feeders near dense shrubs — cardinals prefer cover close by
- Offer a shallow birdbath: cardinals bathe and drink regularly

COLORING GUIDE

ZONE	RECOMMENDED COLOR
Male body	Vivid red — cadmium red
Facial mask	Deep black
Bill	Orange-coral
Female body	Warm sandy brown
Wings & crest ■	Rusty-red highlights
Eye	Dark brown, black ring

COLOR ME!

Northern Cardinal • *Cardinalis cardinalis* • Module 01



MODULE 02 / 40

Black-capped Chickadee

*Poecile atricapillus***QUICK STATS**

Length	4.7 – 5.9 in (12 – 15 cm)
Wingspan	6.3 – 8.3 in (16 – 21 cm)
Weight	0.3 – 0.5 oz (9 – 14 g)
Lifespan	Up to 12 years (avg. 2–3 in wild)

PRESENTATION

The Black-capped Chickadee is one of the most beloved and recognizable backyard birds across northern North America. Bold, curious, and remarkably tolerant of humans, it is often the first species to discover a newly installed feeder. Its cheerful "chick-a-dee-dee-dee" call is one of the most familiar sounds of North American winters. Despite its tiny size, the chickadee is supremely adapted to cold climates, capable of surviving temperatures well below freezing through a combination of extraordinary memory, metabolic flexibility, and dense winter plumage.

HABITAT & RANGE

The Black-capped Chickadee occupies a vast range across the northern tier of North America, from Alaska and the Yukon east to Newfoundland and Nova Scotia, and south through the northern United States including New England, the Great Lakes region, and the Rocky Mountain states down to northern New Mexico. It inhabits deciduous and mixed forests, forest edges, riparian corridors, suburban parks, and wooded gardens. It is a permanent resident throughout its range and does not migrate, relying instead on cached food and physiological adaptations to endure harsh winters.

DIET & FEEDING

- Black-oil sunflower seeds — the top choice at any feeder
- Suet cakes — critical high-energy winter food source
- Nyjer (thistle) seed and hulled sunflower chips
- Peanut pieces and peanut butter mixtures
- Mealworms (live or dried) during breeding season
- Wild: insects, insect eggs, spiders, berries, seeds

BEHAVIOR & SOCIAL LIFE

Chickadees are highly social birds that form stable winter flocks of 6–12 individuals with a clear dominance hierarchy. These flocks often serve as the nucleus for mixed-species foraging groups, attracting nuthatches, woodpeckers, and warblers. Within the flock, dominant birds feed first and claim the best roost sites. The chickadee's "chick-a-dee" alarm call is remarkably sophisticated: the number of "dee" notes appended to the call signals the level of threat, with more notes indicating a more dangerous predator nearby. Researchers have identified it as one of the most complex vocalizations in the animal kingdom.

NESTING & BREEDING

Breeding pairs form in late winter as flocks begin to break up. Chickadees are cavity nesters, excavating their own nest holes in soft rotting wood or using natural cavities and nest boxes. The female lines the cavity with moss, plant fibers, fur, and feathers, creating a deep insulating cup. Clutches typically contain 6–8 white eggs with reddish-brown speckles. Incubation lasts 12–13 days, performed by the female alone while the male brings her food. Both parents feed the nestlings, which fledge at 16 days. Pairs raise one brood per year.

SONG & COMMUNICATION

The chickadee has two primary vocalizations. Its contact and alarm call — the iconic "chick-a-dee-dee-dee" — varies in complexity based on context and threat level. Its territorial song is a clear, whistled two-note "fee-bee" (or "fee-bee-bee"), the first note higher than the second, heard most often on mild late-winter days as males begin to establish breeding territories. Chickadees also produce a variety of soft gargling calls within their flocks.

CONSERVATION STATUS

IUCN Red List:	Least Concern	Population trend:	Stable
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The Black-capped Chickadee remains abundant across its vast range, with an estimated population exceeding 100 million individuals. It adapts well to suburban and urban environments and benefits from nest box programs and winter feeding.

FEEDER TIPS

- Tube feeders or small hopper feeders work best
- Black-oil sunflower seeds are strongly preferred over striped
- Chickadees can be hand-tamed with patience over several weeks
- Place a nest box (1 1/8" hole diameter) near woodland edge
- Suet feeders are especially valued from October through March

COLORING GUIDE

ZONE	RECOMMENDED COLOR
Crown & bib	Deep black
Cheek patches	Pure white
Back & wings	Slate gray
Belly	White to pale gray
Flanks	Buffy-white
Bill	Short, black

COLOR ME!

Black-capped Chickadee • *Poecile atricapillus* • Module 02



MODULE 03 / 40

Blue Jay

*Cyanocitta cristata***QUICK STATS**

Length	9.8 – 11.8 in (25 – 30 cm)
Wingspan	13.4 – 16.9 in (34 – 43 cm)
Weight	2.5 – 3.5 oz (70 – 100 g)
Lifespan	Up to 26 years (avg. 7 in wild)

PRESENTATION

The Blue Jay is one of the most striking and intelligent birds to visit North American gardens. Its bold blue, white, and black plumage makes it instantly recognizable at any feeder. Despite its reputation as a bully — it dominates most other species at feeding stations — the Blue Jay plays a vital ecological role as one of the continent's most important seed dispersers. Highly adaptable and extraordinarily vocal, it produces a wide variety of calls including convincing imitations of hawks, which it uses to clear feeders of competing birds.

HABITAT & RANGE

The Blue Jay is found throughout the eastern half of North America, from southern Canada (Alberta east to Nova Scotia and Newfoundland) south through Florida and west to the edge of the Great Plains. It inhabits a wide range of wooded environments including deciduous and mixed forests, forest edges, suburban neighborhoods, city parks, and gardens with mature oak trees. Northern populations are partially migratory, with large flocks moving south in autumn along coastlines and ridges, though many individuals remain year-round even in cold climates.

DIET & FEEDING

- Acorns — by far the most important food source
- Whole and shelled peanuts — a feeder favorite
- Sunflower seeds (black-oil and striped)
- Corn (cracked or whole kernel)
- Insects, grasshoppers, beetles during summer
- Occasionally: eggs and nestlings of other birds

BEHAVIOR & SOCIAL LIFE

Blue Jays are highly social outside the breeding season, gathering in loose flocks and traveling together through the forest canopy. They are among the most vocal birds in North America, producing a rich repertoire of calls including their signature loud "jay-jay-jay" alarm, musical liquid gurgles, whisper songs, and remarkably accurate mimicry of Red-shouldered and Red-tailed Hawks. This mimicry serves multiple purposes: alerting other jays to hawk presence and, controversially, potentially clearing feeders so the jay can feed undisturbed. Blue Jays have demonstrated problem-solving abilities and tool use in laboratory settings, ranking among the most cognitively advanced birds studied.

NESTING & BREEDING

Blue Jays form monogamous pairs that often mate for life. Both sexes participate in nest building, constructing a bulky open cup of twigs, bark strips, mud, and grass in the fork of a tree or shrub, typically 10–25 feet above ground. Clutches of 2–7 eggs (usually 4–5) are incubated for 17–18 days by both parents. Nestlings are fed by both parents and fledged at 17–21 days. Pairs typically raise one brood per year. Both parents aggressively mob and chase away predators, including owls, hawks, cats, and humans.

SONG & COMMUNICATION

The Blue Jay's vocal repertoire is one of the most varied of any North American bird. Its primary call is a loud, harsh "jay" or "jeer," used as an alarm. It also produces a bell-like "tool-tool" call, soft musical whisper songs audible only at close range, and uncannily accurate imitations of at least three hawk species. Researchers believe jays use hawk mimicry both as a warning system and as a competitive strategy at feeders.

CONSERVATION STATUS

IUCN Red List:	Least Concern	Population trend:	Stable
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Blue Jay populations are stable across their range, with an estimated 45 million individuals in North America. The species has benefited from suburban expansion and the planting of oak trees in urban environments.

FEEDER TIPS

- Platform feeders or large hopper feeders accommodate their size
- Whole peanuts in the shell are irresistible and provide great viewing
- Scatter cracked corn on the ground — jays readily feed at ground level
- Plant oak trees: a single oak can produce thousands of acorns annually
- Expect jays to cache food — they may visit and depart rapidly with full beaks

COLORING GUIDE

ZONE	RECOMMENDED COLOR
Crest & upper wings	Bright blue with black bars
Tail	Blue with black barring, white tips
Face & nape	White with bold black necklace
Breast & belly	White to pale gray
Back	Blue-gray
Bill & legs	Black

COLOR ME!

Blue Jay • *Cyanocitta cristata* • Module 03



MODULE 04 / 40

House Finch

*Haemorhous mexicanus***QUICK STATS**

Length	5.0 – 6.0 in (12.7 – 15.2 cm)
Wingspan	8.0 – 10.0 in (20.3 – 25.4 cm)
Weight	0.6 – 0.9 oz (16 – 27 g)
Lifespan	Up to 11 years (avg. 2–3 in wild)

PRESENTATION

The House Finch is one of the most abundant and widespread songbirds in North America, found from coast to coast in virtually every habitat where humans have settled. Originally native to the western United States and Mexico, it was introduced to the eastern US in the 1940s when illegal pet trade birds were released on Long Island, New York. It has since colonized the entire continent with remarkable speed. The male's rosy-red head and breast make it one of the most colorful regular visitors to backyard feeders, while the streaked brown female is a master of camouflage.

HABITAT & RANGE

Today the House Finch is found across all 48 contiguous United States, southern Canada, and Mexico. It thrives in urban and suburban environments, farmlands, forest edges, and desert scrub. Highly adaptable, it nests on building ledges, in ornamental conifers, hanging baskets, and even inside cacti in the Southwest. Unlike many finches, it does not undertake long migrations, remaining year-round across most of its range.

DIET & FEEDING

- Nyjer (thistle) seed — strongly preferred
- Black-oil sunflower seeds and hulled chips
- White millet scattered on platform feeders
- Wild berries: mulberry, pyracantha, crabapple
- Flower buds and plant shoots in spring
- Weed seeds from dandelion, chickweed, and thistle

BEHAVIOR & SOCIAL LIFE

House Finches are highly gregarious birds that gather in flocks of dozens or even hundreds outside the breeding season. They are cheerful, active, and vocal at feeders, often tolerating close human approach. Males sing a long, rambling warble year-round — one of the few North American songbirds that sings regularly through winter. Fascinatingly, the intensity of the male's red coloration is entirely diet-dependent: birds consuming more carotenoid-rich foods (berries, certain plants) develop brighter red plumage, while poorly-fed males may appear orange or yellow. Females actively choose the reddest males as mates, using color as a reliable indicator of foraging ability and health.

NESTING & BREEDING

House Finches are remarkably flexible nesters, using an extraordinary range of sites including tree cavities, cliff ledges, building eaves, hanging flower baskets, and old nests of other species. The female builds a compact cup of grasses, leaves, and fine plant fibers. Clutches of 2–6 pale blue eggs with dark speckles are incubated for 13–14 days by the female alone. The male feeds her during incubation. Nestlings fledge at 12–19 days. Pairs raise 2–3 broods per year, sometimes beginning a new nest before the previous brood has fully fledged.

SONG & COMMUNICATION

The male House Finch produces one of the most melodious and sustained songs of any common feeder bird — a rich, cheerful warble lasting 3 seconds or more, delivered from a prominent perch. Song is heard year-round but peaks in spring. The call is a sharp "cheep" often given in flight. Flocks maintain contact with soft chirping notes while foraging together.

CONSERVATION STATUS

IUCN Red List:	Least Concern	Population trend:	Stable
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With an estimated population of over 267 million individuals, the House Finch is one of the most numerous birds in North America. However, populations in the eastern US suffered dramatic declines in the 1990s due to a contagious eye disease (Mycoplasmal conjunctivitis) that spread rapidly through feeder populations. Numbers have since partially recovered. If you see a finch with red, swollen, or crusty eyes at your feeder, clean feeders immediately with a 10% bleach solution.

FEEDER TIPS

- Tube feeders with small ports are ideal for finches
- Nyjer seed in a dedicated thistle sock attracts large flocks
- Clean feeders regularly — House Finches are susceptible to eye disease
- Platform feeders with millet attract ground-feeding individuals
- House Finches often feed in large, noisy groups — expect crowds

COLORING GUIDE

ZONE	RECOMMENDED COLOR
Head & breast (male)	Raspberry red to orange-red
Back & wings	Brown with dark streaking
Belly	White with brown streaks
Rump (male)	Reddish-pink
Female (entire)	Plain brown, heavily streaked
Bill	Short, conical, gray-brown

COLOR ME!

House Finch • *Haemorhous mexicanus* • Module 04



MODULE 05 / 40

Ruby-throated Hummingbird

*Archilochus colubris***QUICK STATS**

Length	2.8 – 3.5 in (7 – 9 cm)
Wingspan	3.1 – 4.3 in (8 – 11 cm)
Weight	0.07 – 0.21 oz (2 – 6 g)
Lifespan	Up to 9 years (avg. 3–5 in wild)

PRESENTATION

The Ruby-throated Hummingbird is the only hummingbird species that regularly breeds in eastern North America, and one of the most extraordinary birds on the continent. Weighing less than a nickel, it is a marvel of biological engineering: its wings beat 53 times per second, its heart beats up to 1,200 times per minute during flight, and it is the only bird capable of sustained backwards flight. The male's iridescent ruby-red gorget (throat patch) can appear jet black in poor lighting, then flash brilliant red as the angle of light changes — one of nature's most spectacular optical effects.

HABITAT & RANGE

During summer, the Ruby-throated Hummingbird breeds across the entire eastern half of North America, from Nova Scotia and Manitoba south through Florida and west to the Great Plains. It inhabits forest edges, gardens, orchards, and meadows with abundant flowering plants. Each fall, it undertakes one of the most astonishing migrations in the bird world: a non-stop flight of 500–900 miles across the Gulf of Mexico to its wintering grounds in Mexico and Central America. It returns north each spring from April through May.

DIET & FEEDING

- Flower nectar — trumpetvine, salvia, monarda, impatiens
- Feeder solution: 1 part white sugar to 4 parts water (no dye)
- Tiny insects and spiders — essential protein source
- Tree sap from sapsucker wells (used opportunistically)
- Never use honey, artificial sweeteners, or red dye in feeders

BEHAVIOR & SOCIAL LIFE

Ruby-throated Hummingbirds are among the most aggressively territorial birds in North America relative to their size. Males fiercely defend feeding territories, dive-bombing intruders — including birds many times their size — with surprising ferocity. Outside of courtship and mating, males and females live entirely solitary lives. To survive cold nights and food shortages, hummingbirds enter a state of torpor: their body temperature drops from 40°C to around 18°C, their heart rate falls from 1,200 to just 50 beats per minute, and their breathing nearly stops. This metabolic shutdown can save up to 60% of their nightly energy expenditure.

NESTING & BREEDING

The female alone builds one of nature's most remarkable nests: a walnut-sized cup of plant down and spider silk, camouflaged on the outside with lichens, and expandable as the chicks grow. It is typically placed on a slender downward-sloping branch 10–20 feet above ground. She lays exactly 2 white eggs the size of peas, incubating them alone for 11–14 days. She feeds the nestlings by inserting her bill deep into their throats and regurgitating nectar and insects. Young fledge at 18–22 days. The male plays no role in nesting or chick-rearing.

SONG & COMMUNICATION

Ruby-throated Hummingbirds are not known for melodious song. Their primary vocalizations are sharp, rapid chips and chattering calls used during territorial disputes and pursuit chases. Males perform spectacular U-shaped or pendulum dive displays during courtship, producing a distinctive wing-buzz at the bottom of each arc. The wingbeat itself produces the characteristic humming sound that gives the family its name.

CONSERVATION STATUS

IUCN Red List:	Least Concern	Population trend:	Stable
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With a stable population estimated at nearly 34 million individuals, the Ruby-throated Hummingbird is not currently threatened. However, it faces pressure from habitat loss on its wintering grounds in Central America and from climate change affecting the timing of flower blooms relative to migration. Planting native nectar plants and maintaining clean feeders are the most impactful ways to support local populations.

FEEDER TIPS

- Use a red feeder — no need to add dye to the solution
- Recipe: boil 1 cup white sugar in 4 cups water, cool completely
- Change solution every 2–3 days in hot weather to prevent fermentation
- Clean feeders with hot water and a bottle brush — no soap needed
- Hang feeders by late April in the South, mid-May in the North
- Plant trumpet vine, bee balm, salvia, and cardinal flower nearby

COLORING GUIDE

ZONE	RECOMMENDED COLOR
Back & crown (both)	Iridescent emerald green
Gorget — male throat	Ruby red (may look black)
Breast & belly	White to pale gray
Tail (male)	Forked, dark green-black
Female throat	White, no red gorget
Bill	Long, straight, black

COLOR ME!

Ruby-throated Hummingbird • *Archilochus colubris* • Module 05

